

CHIH-HAO (ANDY) TSAI

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Github Portfolio: <https://github.com/andytsai104/andytsai-robotics-portfolio>

Education

M.S. in Robotics and Autonomous Systems

Arizona State University, Tempe, Arizona, United States

Aug. 2024 – May 2026

GPA: 3.5/4.0

B.S. in Mechanical Engineering

National Taipei University of Technology, Taipei, Taiwan

Sep. 2019 – Jun. 2023

GPA: 3.03/4.0

Professional Experience

BELIV Lab, Arizona State University

Research Assistant

Jun. 2025 – Present

Mesa, Arizona

- Built a **BC+TD3** pedestrian policy in **CARLA** for safety-critical AV testing.
- Designed **BEV + kinematic observations** for continuous pedestrian control.
- Automated **BC training** on **ASU SOL HPC** with **Bash** job scripts.
- Increased collision rate from **0% to 45%** while reducing vehicle clearance.

Publications

- **Chih-Hao Tsai**, H. Guo, and J. Zhao, “Closed-Loop Pedestrian Policy Learning for Safety-Critical Scenario Generation in Autonomous Driving,” *Modeling, Estimation and Control Conference (MECC)*, 2026 (Accepted).

Projects

Consensus-Based Multi-Robot Warehouse System

Aug. 2025 – Dec. 2025

- Configured differential-drive robot models with **2D LiDAR**, **URDF** using **Solidworks**
- Implemented a **ROS 2 + Gazebo** warehouse simulation with **2D LiDAR** robots.
- Integrated allocation results to **Nav2** goal dispatch and task-status monitoring.

Pololu 3pi+ 2040 Embedded Robotics

Aug. 2025 – Dec. 2025

- Integrated IMU, encoders, IR sensors, bump sensors, and differential-drive actuation on a mobile robot.
- Implemented and tuned feedback control for line tracking, hill climbing, and obstacle avoidance.
- Used real-time telemetry to diagnose sensor, motion, and control-performance issues.

Vision-Based Maze Solving & Path Planning with MyCobot Pro 600

Mar. 2025 – Apr. 2025

- Created a **SolidWorks-based URDF digital twin** of a 6-DOF robotic arm for kinematic and motion validation.
- Validated forward/inverse kinematics and joint trajectories in **RViz**, **Gazebo**, **MATLAB**, and **Python**.
- Integrated image-based path planning with physical robot execution through **TCP/IP control**.
- Resolved coordinate-frame, joint-constraint, and trajectory issues during sim-to-real integration.

Robot Kinematics Simulation

Feb. 2025 – Mar. 2025

- Built **SolidWorks** and **URDF** models for Dobot Magician Lite and SCARA robotic arms.
- Derived and validated **forward/inverse kinematics** using **MATLAB** and **Python**.
- Simulated SCARA trajectories and motion-control behavior in **Simulink**, **ROS 2**, and **Gazebo**.

Autonomous Mobile Vehicle and Robotic Arm

Feb. 2022 – Nov. 2022

- Designed and built an autonomous mobile platform with an integrated robotic arm for object relocation.
- Modeled custom components in **SolidWorks** and fabricated prototypes through **3D printing**.
- Integrated the chassis, motors, arm mechanism, camera, and Arduino-based control into a working prototype.
- Developed webcam-based object detection with **TensorFlow**, **OpenCV**, and **Python**.

Technical Skills

- **Mechanical Design:** SolidWorks, 3D CAD modeling, mechanical prototyping, robot component design
- **Robotics & Mechatronics:** robotic arms, mobile robots, URDF/xacro, ROS 2, Gazebo, RViz, sensor and actuator integration
- **Analysis & Control:** forward/inverse kinematics, PID/PI control, motion control, path control, MATLAB, Simulink, sim-to-real validation
- **Programming & Tools:** Python, C/C++, MATLAB, OpenCV, Arduino, Linux, Git, TCP/IP